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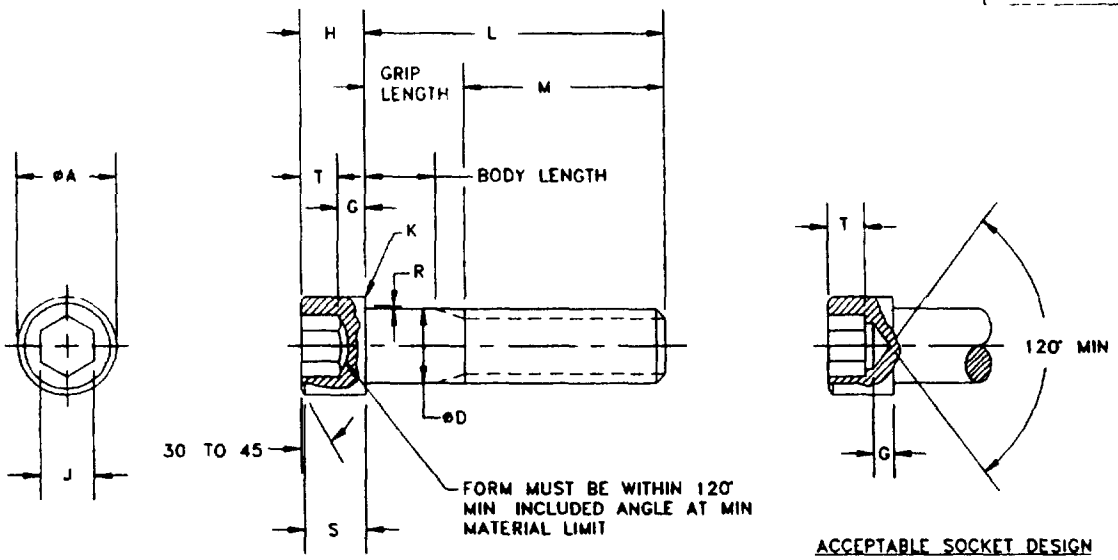


TABLE I DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE			#2 (0860)			#4 (1120)			#6 (1380)		
THREADS PER INCH			56 UNC-3A			40 UNC-3A			32 UNC-3A		
ØD	BODY	MAX	0850			1120			1380		
		MIN	0822			1075			1329		
ØA	HEAD	MAX	1400			1830			2260		
		MIN	1340			1760			2180		
H	HEAD HEIGHT	MAX	0860			1120			1380		
		MIN	0830			1080			1340		
S	HEAD SIDE HEIGHT	MIN	0770			1010			1240		
J	SOCKET WIDTH ACROSS FLATS	MAX	0791			0952			1111		
		MIN	0781			0937			1094		
T	KEY ENGAGEMENT	MIN	0380			0510			0640		
G	WALL THICKNESS	MIN	0280			0380			0440		
K	CHAMFER OR RADIUS	MAX	0030			0050			0050		
R	FILLET RADIUS	MAX	0080			0090			0100		
		MIN	0040			0050			0060		
M	BASIC THREAD LENGTH 1/	REF	6250			7500			7500		
MIN TENSILE STRENGTH LOAD LBS 2/			300			480			730		
L LENGTH	TOLERANCE		DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN
188			1								
250	+ 000		2			9			16		
375	- 030		3			10			17		
500			4			11			18		
625									19		
750									20		

1/ SEE NOTE 2
2/ BASED ON 80,000 PSI MINIMUM TENSILE STRENGTH LOAD POUNDS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2

INCH-POUND

PREPARING ACTIVITY DLA-IS CUSTODIANS ARMY- NAVY-OS AIR FORCE-82 DLA- REVIEW ARMY-MI NAVY-MC SH AIR FORCE-11 NSA-NS PROJECT NUMBER 5305-2081	MILITARY SPECIFICATION SHEET TITLE SCREW, CAP SOCKET HEAD-HEXAGON, CORROSION RESISTANT STEEL, UNC-3A	SPECIFICATION SHEET NUMBER MS16995 27 JUNE 95 REV G SUPERSEDING MS16995F 22 FEB 91 AMSC-N/A FSC-5310

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TABLE I DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			#8 (1640)			#10 (1900)			#1/4 (250)		
THREADS PER INCH			32 UNC-3A			24 UNC-3A			20 UNC-3A		
ØD	BODY	MAX MIN	1640 1585			1900 1840			2500 2435		
ØA	HEAD	MAX MIN	2700 2620			3130 3030			3750 3650		
H	HEAD HEIGHT	MAX MIN	1640 1590			1900 1850			2500 2440		
S	HEAD SIDE HEIGHT	MIN	1480			1710			2250		
J	SOCKET WIDTH ACROSS FLATS	MAX MIN	1426 1406			1587 1562			1900 1875		
T	KEY ENGAGEMENT	MIN	0770			0900			1200		
G	WALL THICKNESS	MIN	0520			0610			0800		
K	CHAMFER OR RADIUS	MAX	0050			0050			0080		
R	FILLET RADIUS	MAX MIN	0120 0070			0140 0090			0140 0090		
M	BASIC THREAD LENGTH 1/	REF	8750			8750			1 0000		
MIN TENSILE STRENGTH LOAD LBS 2/			1 120			1,400			2 540		
L LENGTH	TOLERANCE		DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN
375 500 625 750 875	+ 000 - 030		25 26 27 28 29			35 36 37 38 39			47 48 49 50 51		
1.000			30			40			52		
1 250	+ 000					41	375	167	53		
1 500 1 750 2 000	- 040					42	375	167	54 55 56	500 500 1 000	250 250 750

1/ SEE NOTE 2
2/ BASED ON 80 000 PSI MINIMUM TENSILE STRENGTH LOAD POUNDS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2

PREPARING ACTIVITY DLA-IS CUSTODIANS ARMY- NAVY-OS AIR FORCE 82 DLA REVIEW ARMY-MI NAVY-MC SH AIR FORCE-11 NSA-NS PROJECT NUMBER 5305-2081	MILITARY SPECIFICATION SHEET TITLE SCREW, CAP SOCKET HEAD-HEXAGON, CORROSION RESISTANT STEEL, UNC-3A	SPECIFICATION SHEET NUMBER MS16995 27 JUNE 95 REV G SUPERSEDING MS16995F 22 FEB 91 AMSC N/A FSC-5310

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TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			5/16 (.3125)	3/8 (.375)	1/2 (.500)
THREADS PER INCH			18 UNC-3A	16 UNC-3A	13 UNC-3A
*D	BODY	MAX	.3125	.3750	.5000
		MIN	.3053	.3678	.4919
*A	HEAD	MAX	.4690	.5630	.7500
		MIN	.4570	.5500	.7350
H	HEAD HEIGHT	MAX	.3130	.3750	.5000
		MIN	.3060	.3680	.4920
S	HEAD SIDE HEIGHT	MIN	.2810	.3370	.4500
J	SOCKET WIDTH ACROSS FLATS	MAX	.2530	.3160	.3790
		MIN	.2500	.3125	.3750
T	KEY ENGAGEMENT	MIN	.1510	.1820	.2450
G	WALL THICKNESS	MIN	.1000	.1200	.1600
K	CHAMFER OR RADIUS	MAX	.0080	.0080	.0100
R	FILLET RADIUS	MAX	.0170	.0020	.0260
		MIN	.0120	.0150	.0200
M	BASIC THREAD LENGTH 1/	REF	1.1250	1.2500	1.5000
MIN TENSILE STRENGTH LOAD LBS 2/			4 190	6 200	11 300

L LENGTH	TOLERANCE		DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN	DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN
	SIZE 5/16 (.3125) AND 3/8 (.375)	SIZE 1/2 (.500)									
.375	+.000 -.030	+.000 -.030	61			77					
.500			62			78					
.625			63			79			92		
.750			64			80			93		
.875			65								
1.000			66			81			94		
1.250			67			82			95		
1.500			68			83			96		
1.750	+.000 -.040	+.000 -.060	69	.625	.347	84	.500	.187	97		
2.000			70			85	.500	.187	98		
2.250			71	1.125	.847	86	1.000	.687	99	.750	.365
2.500			72	1.125	.847	87	1.000	.687	100	.750	.365
2.750	+.000 -.060	+.000 -.080							101	.750	.365
3.000									102	1.500	1.115

1/ SEE NOTE 2
2/ BASED ON 80,000 PSI MINIMUM TENSILE STRENGTH. LOAD POUNDS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- NAVY-OS
AIR FORCE-82 DLA-
REVIEW ARMY-MI NAVY-MC SH AIR FORCE-11 NSA-NS
PROJECT NUMBER 5305-2081

MILITARY SPECIFICATION SHEET

TITLE

SCREW, CAP SOCKET HEAD-HEXAGON CORROSION RESISTANT STEEL UNC-3A

SPECIFICATION SHEET NUMBER
MS16995 27 JUNE 95 REV G

SUPERSEDING
MS16995F 22 FEB 91

AMSC- N/A FSC-5310

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TABLE I DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			5/8 (625)		
THREADS PER INCH			11 UNC-3A		
øD BODY		MAX MIN	6250 6163		
øA HEAD		MAX MIN	9360 9210		
H HEAD HEIGHT		MAX MIN	6250 6160		
S HEAD SIDE HEIGHT		MIN	5620		
J SOCKET WIDTH ACROSS FLATS		MAX MIN	5050 5000		
T KEY ENGAGEMENT		MIN	3070		
G WALL THICKNESS		MIN	2000		
K CHAMFER OR RADIUS		MAX	0100		
R FILLET RADIUS		MAX MIN	0320 0240		
M BASIC THREAD LENGTH 1/		REF	1 7500		
MIN TENSILE STRENGTH LOAD LBS 2/			18 100		
L LENGTH	TOLERANCE		DASH NO	GRIP LENGTH MAX	BODY LENGTH MIN
1 000	+ 000 - 030		107		
1 250 1 500 1 750 2 000	+ 000 - 060		108 109 110 111		
2 250			112		
2 500 2 750 3 000	+ 000, - 080		113 114 115	750 750 750	295 295 295

1/ SEE NOTE 2

2/ BASED ON 80 000 PSI MINIMUM TENSILE STRENGTH LOAD POUNDS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2

REQUIREMENTS

1 MATERIAL. CORROSION RESISTANT STEEL IN ACCORDANCE WITH PROCUREMENT SPECIFICATION OR TYPE 303 (UNS S30300) EQUAL TO OR INTERCHANGEABLE WITH 16 18 CHROMIUM NICKEL ALLOY STEEL DEVELOPED FOR COLD HEADING

2 CLEANED, DESCALED AND PASSIVATED. SCREWS SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH QQ-P-35

3 PROTECTIVE COATING AND SURFACE TREATMENT. SCREWS SHALL BE BLACK OXIDE COATED IN ACCORDANCE WITH MIL-C-13924, CLASS 4 (WHEN BLACK OXIDE COATING IS REQUIRED, THE DASH NUMBER SHALL BE FOLLOWED BY A CODE LETTER "B")

4 MAGNETIC PERMEABILITY. SCREWS SHALL HAVE A MAGNETIC PERMEABILITY OF 210 MAXIMUM (AIR = 1 0) FOR A FIELD STRENGTH OF H = 200 OERSTEDS USING A MAGNETIC INDICATOR IN ACCORDANCE WITH MIL-I-17214

(G)5 THREADS. ACCEPTABILITY OF SCREW THREADS FOR SIZES #2 (0860) THRU #10 (190) NOMINAL DIAMETERS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21 ACCEPTABILITY OF SCREW THREADS 250 AND LARGER SHALL BE IN ACCORDANCE WITH FED-STD-H28/20 SYSTEM 22 SCREWS ABOVE THE HEAVY LINE SHALL HAVE COMPLETE (FULL FORM) THREADS EXTENDED TO WITHIN TWO (2) THREADS OF THE HEAD AS MEASURED WITH A THREAD RING GAGE

6 SOURCE IDENTIFICATION MARK. SOURCE IDENTIFICATION MARK SHALL BE IN ACCORDANCE WITH PROCUREMENT SPECIFICATION

7 HEAD STYLE. HEAD OF SCREW MAY BE KNURLED OR NOT KNURLED AT THE OPTION OF THE MANUFACTURER TYPE OF KNURLING OPTIONAL

PREPARING ACTIVITY DLA-IS CUSTODIANS ARMY- NAVY-OS AIR FORCE-82 DLA REVIEW ARMY-MI NAVY-MC,SH AIR FORCE-11 NSA-NS PROJECT NUMBER 5305-2081	MILITARY SPECIFICATION SHEET TITLE SCREW CAP SOCKET HEAD-HEXAGON CORROSION RESISTANT STEEL, UNC-3A	SPECIFICATION SHEET NUMBER MS16995 27 JUNE 95 REV G SUPERSEDING MS16995F 22 FEB 91 AMSC-N/A FSC-5310

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8 PART NUMBER. THE PART NUMBER SHALL CONSIST OF THE BASIC MS SHEET NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND A CODE LETTER "B" FOR BLACK OXIDE IF APPLICABLE

EXAMPLE MS16995-1

DASH NUMBER
BASIC MS SHEET NUMBER

MS16995-1 INDICATES - SCREW CAP SOCKET HEAD, HEXAGON CORROSION RESISTANT STEEL, 0860 56 UNC-3A NOMINAL THREAD SIZE 188 LENGTH PASSIVATED

MS16995-1B INDICATES - SCREW, CAP SOCKET HEAD HEXAGON CORROSION RESISTANT STEEL 0860-56 UNC-3A, NOMINAL THREAD SIZE 188 LENGTH BLACK OXIDE COATING

NOTES

1 ALL DIMENSIONS ARE IN INCHES
2 THE BASIC THREAD LENGTH "W" (TABLE I) IS A REFERENCED VALUE CONTROLLED BY THE GRIP LENGTH AND "L" LENGTH AS SPECIFIED
3 MS16995 SUPERSEDES MS35461 IN PART
4 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE
5 UNLESS OTHERWISE SPECIFIED ISSUES OF REFERENCED DOCUMENTS ARE IN EFFECT AT THE TIME OF SOLICITATION

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THE SCREWS COVERED BY DASH NUMBERS GIVEN IN MS35461 THAT ARE REPLACED BY APPLICABLE DASH NUMBERS IN THIS DOCUMENT ARE CANCELLED AFTER 5 APRIL 62. USE THE DASH NUMBERS GIVEN IN THE PRECEDING SHEETS THE CANCELLED SCREWS CANNOT ALWAYS REPLACE THE NEW SCREWS AND SHOULD BE USED UNTIL EXISTING STOCKS ARE DEPLETED USE ONLY THE NEW SCREWS FOR DESIGN AND REPLACEMENT REPLACEMENT SHALL BE IN ACCORDANCE WITH THIS TABLE INTERCHANGEABILITY INFORMATION REGARDING PART NUMBERS ON SUPERSEDED DRAWINGS ARE ALSO CNTAINED IN THIS TABLE

INTERCHANGEABILITY TABLE

PART NUMBERS		PART NUMBERS		PART NUMBERS	
CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING
MS35461	MS16995	MS35461	MS16995	MS35461	MS16995
DASH NUMBER	DASH NUMBER	DASH NUMBER	DASH NUMBER	DASH NUMBER	DASH NUMBER
1	25	23	49	54	83
2	26	24	50	55	84
3	27	25	51	56	85
4	28	26	52	57	86
5	29	27	53	58	87
6	30	28	54	62	92
11	35	29	55	63	93
12	36	30	56	64	94
13	37	48	77	65	95
14	38	49	78	66	96
15	39	50	79	67	97
16	40	51	80	68	98
21	47	52	81	69	99
22	48	53	82	70	100

SCREWS COVERED BY THE FOLLOWING PART NUMBERS ARE INACTIVE FOR NEW DESIGN AND ENGINEERING THEY SHALL BE USED FOR MAINTAENANCE OF EXISTING EQUIPMENT TO REPLACE THE 5/16 INCH SIZE SCREWS USED IN COUNTERBORED HOLES OR OTHER CLOSE APPLICATIONS

PART NUMBERS
INACTIVE
MS35461
DASH NUMBER
33
34
35
36
37
38
39
40
41
42
43
44

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- NAVY-OS
AIR FORCE- 82 DLA-
REVIEW ARMY-MI NAVY-MC SH AIR FORCE-11
NSA-NS
PROJECT NUMBER 5305-2081

MILITARY SPECIFICATION SHEET

TITLE

SCREW, CAP SOCKET HEAD-HEXAGON, CORROSION RESISTANT STEEL UNC-3A

SPECIFICATION SHEET NUMBER
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SUPERSEDING
MS16995F 22 FEB 91
AMSC- N/A FSC -5310

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